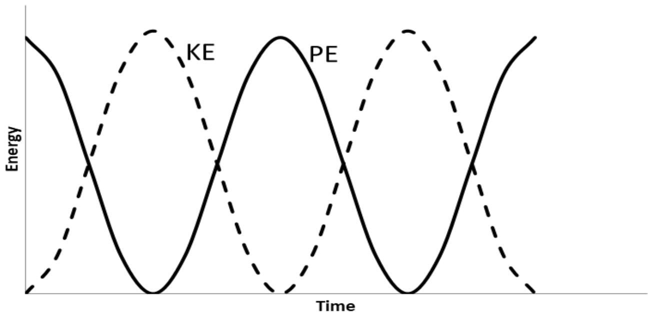
Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(iii)	I.	$T = \frac{2\pi}{\omega} = \frac{2\pi}{7.7}$ = 0.82 [s]		1		1	1	
			II.	Substitution: $T = 2\pi\sqrt{\frac{m}{k}}$ so $0.82 = 2\pi\sqrt{\frac{0.30}{k}}$ **or** $ma = kx$ (1) k = 17.8 N m^{-1} **unit mark** (1)	1	1		2	2	
			III.	Substitution: $v = wA = 7.7 \times 0.05$ (1) if sinωt or cosωt used then the angle must be correct v = 0.39 [m s^{-1}] (1) **Alternatives for the first mark:** Substituting into $ma = kx$ e.g. $0.3 \times 3 = k \times 0.05$ (1) **OR** substituting into $\omega = \sqrt{\frac{k}{m}}$ e.g. $7.8 = \sqrt{\frac{k}{0.3}}$ (1)	1	1		2	2	
		(iv)		 Shape and start point of PE sketch (1) Shape KE sketch and starting at zero (1) KE out of phase with PE and same size peaks and one complete cycle i.e. 2 KE peaks (1)		3		3	1	